Homework 7

1. Convex or Concave

Determine whether the following functions are convex, strictly convex, concave, strictly concave, both or neither.

(a) $f(x) = \mathrm{e}^x - 1$ on $\mathbb{R}$.

Solution: This is strictly convex since $\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}(x) = \mathrm{e}^x > 0$ for all $x \in \mathbb{R}$.

(b) $f(x_1, x_2) = x_1 x_2$ on $\mathbb{R}^2_{++}$, i.e., when $x_1 > 0$ and $x_2 > 0$.

Solution: $f(x_1, x_2) = x_1 x_2$ on $\mathbb{R}^2_{++}$.

This is neither convex nor concave. The Hessian of f is

$$\nabla^2 f(x) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{1}$$

which has eigenvalues ± 1 which implies the Hessian is neither positive semidefinite nor negative semidefinite.

(c) The log-likelihood of a set of points $\{x_1, \ldots, x_n\}$ that are normally distributed with mean μ and finite variance $\sigma > 0$ is given by:

$$f(\mu, \sigma) = n \log\left(\frac{1}{\sqrt{2\pi}\sigma}\right) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 \tag{2}$$

i. Show that if we view the log likelihood for fixed σ as a function of the mean, i.e

$$g(\mu) = n \log\left(\frac{1}{\sqrt{2\pi}\sigma}\right) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 \tag{3}$$

then g is strictly concave (equivalently, we say f is strictly concave in μ).

Solution: For $g(\mu)$ we have,

$$\nabla g(\mu) = \sum_{i=1}^{n} \frac{x_i - \mu}{\sigma^2} \tag{4}$$

$$\nabla^2 g(\mu) = -\frac{n}{\sigma^2} < 0. \tag{5}$$

Since σ is finite, g is strictly concave (equivalently f is strictly concave in μ).

ii. Show that if we view the log likelihood for fixed μ as a function of the inverse of the variance, i.e

$$h(z) = n \log\left(\frac{\sqrt{z}}{\sqrt{2\pi}}\right) - \frac{z}{2} \sum_{i=1}^{n} (x_i - \mu)^2 \tag{6}$$

then h is strictly concave (equivalently, we say f is strictly concave in $z = \frac{1}{\sigma^2}$). Note that we have used the dummy variable z to denote $\frac{1}{\sigma^2}$.

Solution: For $h(z)$ we have,

$$\nabla h(z) = \frac{n}{2z} - \sum_{i=1}^{n} \frac{(x_i - \mu)^2}{2} \tag{7}$$

$$\nabla^2 h(z) = -\frac{n}{2z^2} < 0. \tag{8}$$

Since z^2 is finite ($\sigma > 0$), h is strictly concave (equivalently f is strictly concave in σ^2).

iii. Show that f is not jointly concave in $\mu, \frac{1}{\sigma^2}$. *HINT: We say a function $w(\vec{x}, \vec{y})$ with $\vec{x} \in \mathbb{R}^m$ and $\vec{y} \in \mathbb{R}^n$ is jointly convex if*

$$w\left(\lambda(\vec{x}_1, \vec{y}_1) + (1-\lambda)(\vec{x}_2, \vec{y}_2)\right) \leq \lambda w((\vec{x}_1, \vec{y}_1)) + (1-\lambda) w((\vec{x}_2, \vec{y}_2)). \tag{9}$$

This is the same as letting $\vec{z} = (\vec{x}, \vec{y})$ and saying f is convex in $\vec{z}$. We can define joint concavity in a similar fashion by reversing the inequalities.

Solution: For $f(\mu, \frac{1}{\sigma^2})$, we find the second order partial derivatives and stack them in the Hessian. We have,

$$\nabla^2 f(\mu, \frac{1}{\sigma^2}) = \begin{bmatrix} -\frac{n}{\sigma^2} & \sum_{i=1}^{n}(x_i - \mu) \\ \sum_{i=1}^{n}(x_i - \mu) & -\frac{n\sigma^4}{2} \end{bmatrix}. \tag{10}$$

The determinant of the Hessian is given by,

$$\det\left(\nabla^2 f\right) = \frac{n^2\sigma^2}{2} - (\sum_{i=1}^{n}(x_i - \mu))^2. \tag{11}$$

and the trace of the Hessian is given by,

$$\operatorname{tr}\left(\nabla^2 f\right) = -\frac{n}{\sigma^2} - \frac{n\sigma^4}{2} < 0. \tag{12}$$

Note that the trace is the sum of the eigenvalues, and the determinant is the product of the eigenvalues. Since the trace is always negative, if the determinant is negative it must imply that one eigenvalue is positive and another is negative; that is, we have f is neither convex nor concave. It is easy to see that $\det\left(\nabla^2 f\right)$ can sometimes be negative – for example, if we choose σ^2 to be close to zero and μ away from x_i, the second negative term dominates and make $\det\left(\nabla^2 f\right) \leq 0$.

Aside: Note however, in the maximum likelihood estimates, the Hessian is negative semi-definite implying that locally the function is concave. More concretely, at

$$\widehat{\mu} = \frac{1}{n}\sum_{i=1}^{n}, \quad \widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \widehat{\mu})^2 \tag{13}$$

we have $\nabla^2 f(\widehat{\mu}, 1/\widehat{\sigma}^2) \preceq 0$

(d) $f(x) = \log(1 + \mathrm{e}^x)$. Compare this to $h(x) = \frac{1}{1+\mathrm{e}^x}$, is $h(x)$ convex or concave?

Solution: We will do this by verifying the second order sufficient conditions for convexity. We have the derivatives of f can be computed using the chain rule as follows:

$$f'(x) = \frac{\partial f}{\partial x}(x) = \frac{\mathrm{e}^x}{1 + \mathrm{e}^x} \tag{14}$$

$$f''(x) = \frac{\partial^2 f}{\partial x^2}(x) = \frac{\mathrm{e}^x}{(1 + \mathrm{e}^x)^2} > 0. \tag{15}$$

Since we have $f''(x) > 0$ for all x, we conclude that the function f is convex.

For $h(x) = (1 + e^x)^{-1}$, we compute the second derivative to check for convexity:

$$h'(x) = -\frac{e^x}{(1 + e^x)^2} \tag{16}$$

$$h''(x) = \frac{e^x(e^x - 1)}{(1 + e^x)^3}. \tag{17}$$

The sign of $h''(x)$ depends on $(e^x - 1)$. Thus, $h(x)$ is concave for $x < 0$ (where $h'' < 0$) and convex for $x > 0$ (where $h'' > 0$). Overall, the function is neither purely convex nor concave.

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2. Convex and strictly convex functions

(a) Recall that a function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be strictly convex if it satisfies Jensen's inequality with strict inequality, i.e., $\forall \vec{x} \neq \vec{y} \in \mathbb{R}^n$ and $\forall t \in (0, 1)$, we have

$$f(t\vec{x} + (1-t)\vec{y}) < tf(\vec{x}) + (1-t)f(\vec{y})$$

Show that for a strictly convex function $f : \mathbb{R}^n \to \mathbb{R}$, the problem

$$\min_{\vec{x} \in \mathbb{R}^n} f(\vec{x}) \tag{18}$$

has at most one solution.

HINT: Try to argue by contradiction assuming that there are two solutions $\vec{x}_1, \vec{x}_2$ which achieve the minimum value. Argue that using these two points you can find another point in $\mathbb{R}^n$ with strictly smaller function value.

Solution: Assume that $f^* = \min_{\vec{x} \in \mathbb{R}^n} f(\vec{x})$ has at least two different optimal solutions $\vec{x}_1, \vec{x}_2 \in \mathbb{R}^n$. Hence $f^* = f(\vec{x}_1) = f(\vec{x}_2)$. Consider $\vec{z} = (\vec{x}_1 + \vec{x}_2)/2$.

$$\begin{aligned} f(\vec{z}) &= f((\vec{x}_1 + \vec{x}_2)/2) \\ &< (f(\vec{x}_1) + f(\vec{x}_2))/2 \\ &= f^* \end{aligned}$$

where the inequality follows from strict convexity. Hence we've shown that $\vec{z}$ has functional value strictly smaller than f^* and hence f^* was not the optimal value giving us a contradiction.

(b) Prove that for all convex optimization problems $\min_{\vec{x} \in \mathcal{X}} f(\vec{x})$, where f is a convex function and $\mathcal{X}$ is a convex set, all local minima are global minima. You may not assume that f is differentiable.

HINT: Start with assuming $\vec{x}^\star$ is a local minimum that is not global, and $\vec{\tilde{x}}$ is a global minimum. Use the definition of the convexity of a function to prove by contradiction.

Solution: To arrive at a contradiction, suppose $\vec{x}^\star$ is a local minimum that is not global. Let $\vec{\tilde{x}}$ be a global minimum. Thus we have $f(\vec{\tilde{x}}) < f(\vec{x}^\star)$. Then by convexity $\lambda\vec{x}^\star + (1-\lambda)\vec{\tilde{x}} \in \mathcal{X}$ and hence

$$\begin{aligned} f(\lambda\vec{x}^\star + (1-\lambda)\vec{\tilde{x}}) &\leq \lambda f(\vec{x}^\star) + (1-\lambda)f(\vec{\tilde{x}}) \\ &< \lambda f(\vec{x}^\star) + (1-\lambda)f(\vec{x}^\star) \\ &= f(\vec{x}^\star) \end{aligned}$$

Thus, for all $\lambda \in [0, 1)$, we have $f(\lambda\vec{x}^\star + (1-\lambda)\vec{\tilde{x}}) < f(\vec{x}^\star)$. Then we can make $\lambda\vec{x}^\star + (1-\lambda)\vec{\tilde{x}}$ arbitrarily close to $\vec{x}^\star$, contradicting that $\vec{x}^\star$ is a local minimum.

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3. Quadratic inequalities

Consider the set S defined by the following inequalities:

$$(x_1 \geq -x_2 + 1 \text{ and } x_1 \leq 0) \text{ or } (x_1 \leq -x_2 + 1 \text{ and } x_1 \geq 0)\,. \tag{19}$$

To be more precise,

$$S_1 = \{\vec{x} \in \mathbb{R}^2 \mid x_1 \geq -x_2 + 1, x_1 \leq 0\} \tag{20}$$

$$S_2 = \{\vec{x} \in \mathbb{R}^2 \mid x_1 \leq -x_2 + 1, x_1 \geq 0\} \tag{21}$$

$$S = S_1 \cup S_2. \tag{22}$$

(a) Draw the set S. Is it convex?

Solution:

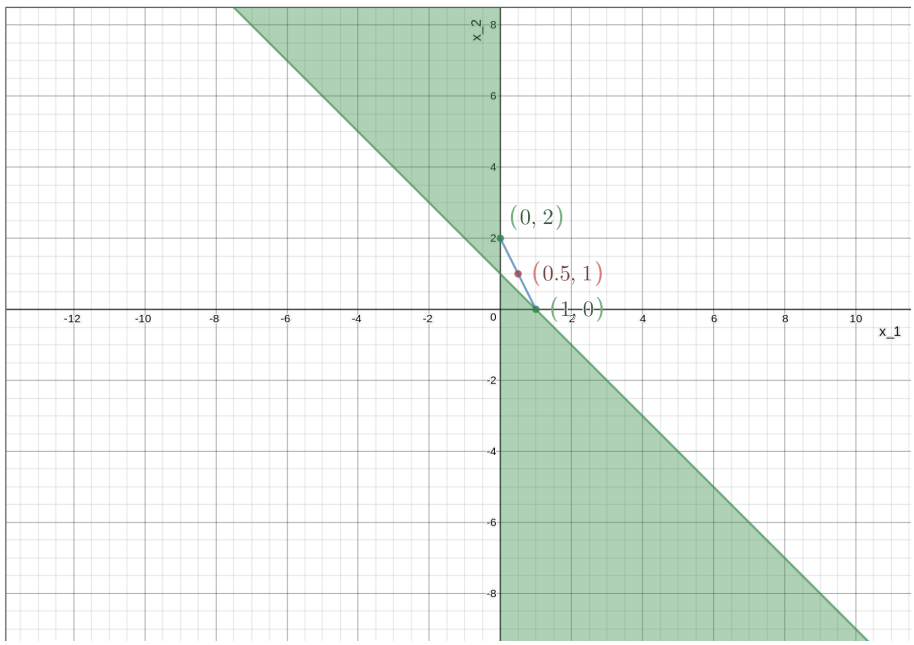

Figure 1: Set S

The set S as shown in Fig. 1 is not convex. We can prove this by counterexample. $(0, 2)$ and $(1, 0)$ both belong to the set, but the midpoint $(1/2, 1)$ does not.

(b) Show that the set S, can be described as a single quadratic inequality of the form $q(\vec{x}) = \vec{x}^\top A\vec{x} + 2\vec{b}^\top \vec{x} + c \leq 0$, for matrix $A = A^\top \in \mathbb{R}^{2\times 2}$, $\vec{b} \in \mathbb{R}^2$ and $c \in \mathbb{R}$ i.e S can be written as $S = \{\vec{x} \in \mathbb{R}^2 \mid q(\vec{x}) \leq 0\}$). Find $A, \vec{b}, c$.

Hint: Can you combine the constraints to make one quadratic constraint?

Solution: Within set S, $x_1 + x_2 - 1 \geq 0$ when $x_1 \leq 0$ and $x_1 + x_2 - 1 \leq 0$ when $x_1 \geq 0$. It follows that $q(\vec{x}) = x_1(x_1 + x_2 - 1) \leq 0$ if and only if it is in the set. Expressing $q(\vec{x})$ in the desired form:

$$q(\vec{x}) = x_1^2 + x_1x_2 - x_1 = \vec{x}^\top A\vec{x} + 2\vec{b}^\top \vec{x} + c$$

where

$$A = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 0 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} -1/2 \\ 0 \end{bmatrix}, \quad c = 0.$$

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(c) The convex hull of a set $A \subseteq \mathbb{R}^n$ is the set of all convex combinations of points in A, i.e.,

$$\text{conv}(A) = \left\{ \sum_{i=1}^{k} \theta_i \vec{x}_i \;\middle|\; k \in \mathbb{N}, \theta_1, \ldots, \theta_k \geq 0, \sum_{i=1}^{k} \theta_i = 1, \vec{x}_1, \ldots, \vec{x}_k \in A \right\}. \tag{23}$$

What is the convex hull of S?

Solution: The convex hull of the set is the whole space, $\mathbb{R}^2$. To see this note than any point $z = (z_1, z_2) \in \mathbb{R}^2$ can be written as $z = \frac{x+y}{2}$ with $x, y \in S$ as follows:
$x = (2z_1, 1 - 2z_1), y = (0, 2(z_1 + z_2) - 1)$.

(d) We will now consider some convex optimization problems over S_1 that illustrate the role of the constraints in the optimization problem. For each of the following optimization problems find the optimal point, $\vec{x}^*$. Describe the constraints that are active in attaining the optimal value. *Hint: Suppose that there exists a point $\vec{x}$ such that $\nabla f(\vec{x}) = 0$. From the first order characterization of a convex function $\vec{x}$ would be an optimum value for f subject to no constraints. If $\vec{x}$ is not in the constraint set S_1, then the optimum point must be on the boundary of the set, i.e. it satisfies at least one of the constraints defining S_1 with equality.*

i. Minimize $f(\vec{x}) = (x_1 + 1)^2 + (x_2 - 3)^2$ subject to $\vec{x} \in S_1$.

Solution: We first compute the unconstrained optimal value of f. Notice that f is a convex function. Therefore, we can compute its optimal value by computing its gradient and setting it to 0. Doing so, we obtain the optimal value of f to be 0 attained at the point $\vec{x}^* = (-1, 3)$. Now, since $\vec{x}^* \in S_1$, $\vec{x}^*$ is the solution to the constrained optimization problem as well.

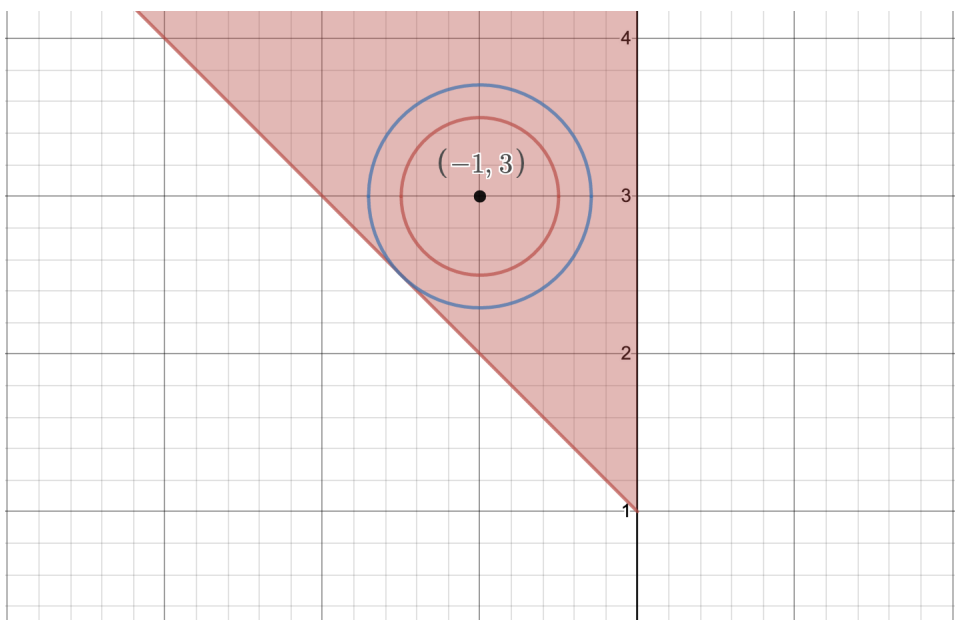

Figure 2: This figure illustrates the position of the optimum, $x^* = (-1, 3)$, and the level sets of the objective function, f, which are concentric circles around x^*.

ii. Minimize $f(\vec{x}) = (x_1 + 2)^2 + (x_2 - 2)^2$ subject to $\vec{x} \in S_1$.

Solution: Proceeding as in the proof for the previous problem, we first find the solution to the unconstrained optimization problem. We get that the unconstrained problem is minimized at the point $\vec{x}_u^* = (-2, 2)$. However, this point is not in the feasible set, S_1. Therefore, the true optimum, $\vec{x}^*$, has

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one or more constraints active. Now, we will attempt to solve the problem with one active constraint. Suppose the one active constraint is $x_1 \geq -x_2 + 1$. Since this constraint is active, we must try and minimize $f(\vec{x})$ subject to $\vec{x}$ satisfying $x_1 = -x_2 + 1$. Note that any point on this line can be written in the form $(0, 1) + \alpha(-1, 1)$. Now consider the function, $g(\alpha)$:

$$g(\alpha) = f((0, 1) + \alpha(-1, 1)) = (\alpha - 2)^2 + (\alpha - 1)^2.$$

Note that the function, $f(\alpha)$, is convex in α. Therefore, we can minimize $g(\alpha)$ by taking its derivative and setting it to 0. By doing this, we get that $\alpha = 3/2$ is the unique minimizer of $g(\alpha)$. Therefore, the minimizer of f subject to $x_1 = -x_2 + 1$ is the point $(-3/2, 5/2)$, and the function value is 0.5. Similarly, the minimizer of f assuming the second constraint, $x_1 \leq 0$, is active is obtained at the point $(0, 2)$, and the function value at this point is 4, which is higher than the value at $(-3/2, 5/2)$. The final possibility is that both constraints are active. However, the optimal value of f subject to both constraints being active will be greater than the value of f obtained at $(-3/2, 5/2)$ which is in S_1. Therefore, we get that $f(\vec{x})$ is minimized at the point $\vec{x}^* = (-3/2, 5/2)$ subject to $\vec{x} \in S_1$. There is one active constraint at $\vec{x}^*$.

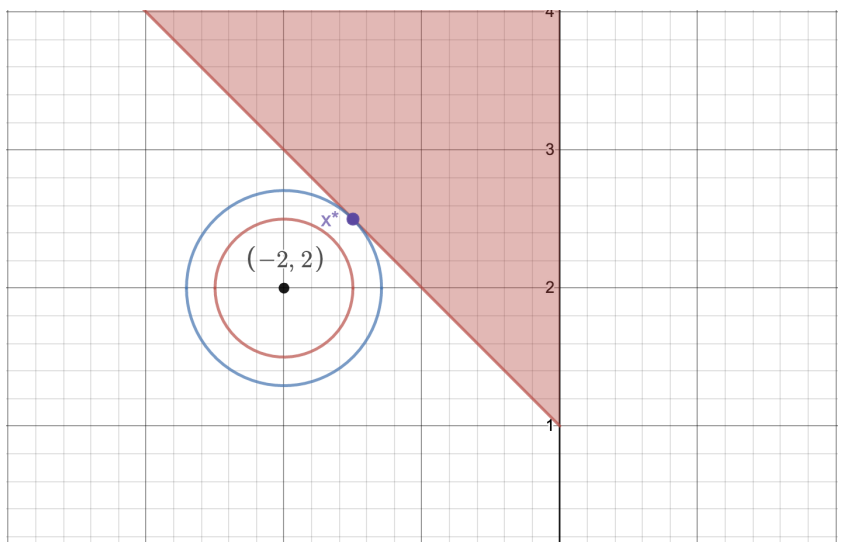

Figure 3: This figure illustrates the position of the optimum, $x^* = (-3/2, 5/2)$, and the level sets of the objective function, f, which are concentric circles around $(-2, 2)$. Note that in this case, the unconstrained optimum does not lie in the set, S_1 and the optimal point lies on the boundary of one of the constraints.

iii. Minimize $f(\vec{x}) = x_1^2 + x_2^2$ subject to $\vec{x} \in S_1$.

Solution: Proceeding as before, we first check the case where 0 constraints are active. However, the unconstrained minimizer of f is $(0, 0)$ which is not in S_1. Now, we check the cases where one of the constraints is active. Assume that the constraint $x_1 \leq 0$ is active. In this case the optimizer is again obtained at the point $(0, 0)$ which is not in S_1. We then consider the case where the constraint $x_1 \geq -x_2 + 1$ is active. As before, we define the function, $g(\alpha)$ as:

$$g(\alpha) = f((0, 1) + \alpha(-1, 1)) = \alpha^2 + (\alpha + 1)^2.$$

By optimizing over α by setting its gradient with respect to α and setting it to 0, we get the optimal setting of α is $-1/2$. However, note that the point $(1/2, 1/2)$ does not belong to S_1 either. Therefore,

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the only remaining possibility is the possibility that both constraints are active. This can happen solely at the point $(0, 1)$. At this point, the value of the function f is 1, the optimizer $\vec{x}^* = (0, 1)$ and both constraints are active at $\vec{x}^*$.

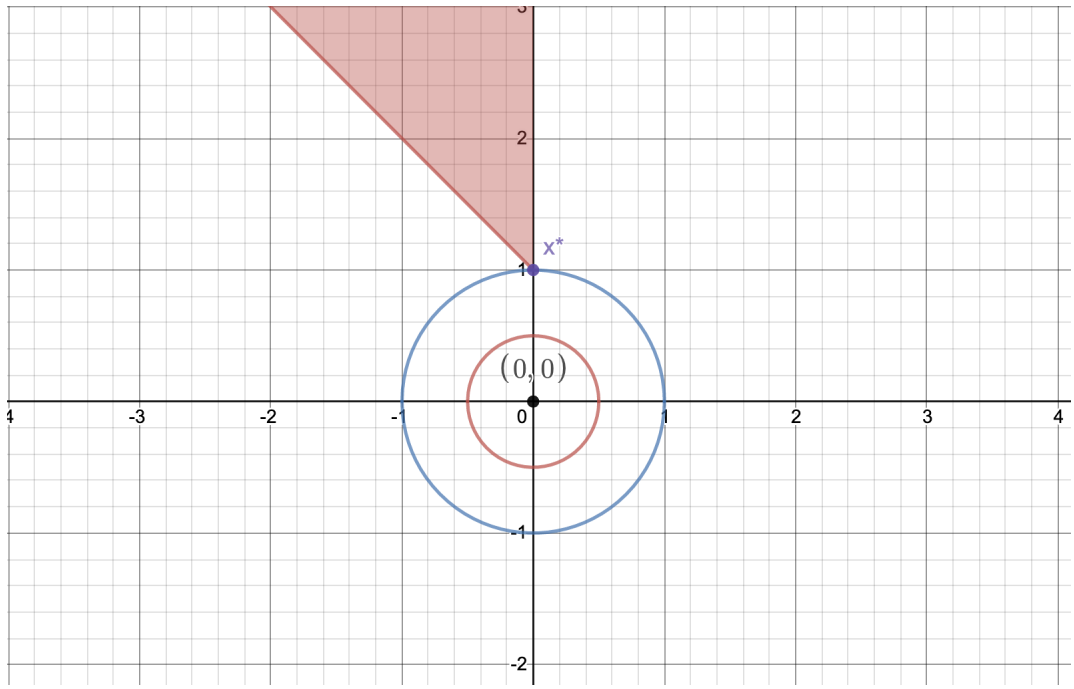

Figure 4: This figure illustrates the position of the optimum, $x^* = (0, 1)$, and the level sets of the objective function, f, which are concentric circles around $(0, 0)$. Note that in this case, the unconstrained optimum does not lie in the set, S_1 and the optimal point lies on the boundary of *both* of the constraints.

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4. Quadratics and Least Squares

In this question, we will see that every least squares problem can be considered as minimization of a quadratic cost function; whereas not every quadratic minimization problem corresponds to a least-squares problem. To begin with, consider the quadratic function, $f : \mathbb{R}^2 \to \mathbb{R}$ given by:

$$f(\vec{w}) = \vec{w}^\top A\vec{w} - 2\vec{b}^\top \vec{w} + c \tag{24}$$

where $A \in \mathbb{S}^2_+$ (set of symmetric positive semidefinite matrices in $\mathbb{R}^{2\times 2}$), $\vec{b} \in \mathbb{R}^2$ and $c \in \mathbb{R}$.

(a) Assume $c = 0$, and assume that setting $\nabla f(\vec{w}) = 0$ allows us to find the unique minimizer. Give a concrete example of a matrix $A \succ 0$ and a vector $\vec{b}$ such that the point $\vec{w}^\star = \begin{bmatrix} -1 & 1 \end{bmatrix}^\top$ is the unique minimizer of the quadratic function $f(\vec{w})$.

Solution: First, let $A \succ 0$. Now, by taking the gradient of $f(\vec{w})$ and setting it to zero, we get:

$$\nabla f(\vec{w}^\star) = 2A\vec{w}^\star - 2b = 0. \tag{25}$$

Since A is positive definite, it is invertible and therefore, the above minimizer is unique. Concretely, let $A = I$. By setting the gradient to zero, we obtain

$$\nabla f(\vec{w}^\star) = (A + A^\top)\vec{w}^\star - 2\vec{b} = 0 \implies \vec{w}^\star = \vec{b}. \tag{26}$$

Then $\vec{w}^\star = [-1\ 1]^\top$ is the unique minimizer if $\vec{b} = [-1\ 1]^\top$ and $A = I$.

(b) Assume $c = 0$. Give a concrete example of a matrix $A \succeq 0$, and a vector $\vec{b}$ such that the quadratic function $f(\vec{w})$ has infinitely many minimizers and all of them lie on the line $w_1 + w_2 = 0$.

HINT: Take the gradient of the expression and set it to zero. What needs to be true for there to be infinitely many solutions to the equation?

Solution: Since $A \in \mathbb{R}^{2\times 2}$ is positive semidefinite, setting gradient to zero shows us that each minimizer $\vec{w}^\star$ satisfies

$$\nabla f(\vec{w}^\star) = (A + A^\top)\vec{w}^\star - 2\vec{b} = 2A\vec{w}^\star - 2\vec{b} = 0. \tag{27}$$

In order to have infinitely many solutions, the positive semidefinite matrix A cannot have full rank. Since $A \in \mathbb{S}^2_+$, this amounts to A having rank at most 1. In other words, there must exist a vector $\vec{v} \in \mathbb{R}^2$ such that $A = \vec{v}\vec{v}^\top$. By setting $A = \vec{v}\vec{v}^\top$, each minimizer $\vec{w}^\star$ should satisfy

$$A\vec{w}^\star - \vec{b} = \vec{v}\vec{v}^\top \vec{w}^\star - \vec{b} = 0. \tag{28}$$

Note that each point on the line

$$\mathcal{L} = \{\vec{w} \in \mathbb{R}^2 : w_1 + w_2 = 0\} = \left\{ \alpha \begin{bmatrix} 1 \\ -1 \end{bmatrix} : \alpha \in \mathbb{R} \right\} \tag{29}$$

is a minimizer of f. Along with (28), this implies that

$$\vec{v}\vec{v}^\top \left(\alpha \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) - \vec{b} = 0 \quad \forall \alpha \in \mathbb{R}. \tag{30}$$

This is satisfied only when $\vec{b} = 0$ and $\vec{v} \perp [-1\ 1]^\top$. Choosing $\vec{v} = [1\ 1]^\top$, we have

$$A = \beta \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad \vec{b} = 0 \quad \text{for some } \beta > 0. \tag{31}$$

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(c) Assume $c = 0$. Let $\vec{w} = \begin{bmatrix} 1 & 0 \end{bmatrix}^\top$. Give a concrete example of a **non-zero** matrix $A \succeq 0$ and a vector $\vec{b}$ such that the quadratic function $f(\alpha\vec{w})$ tends to $-\infty$ as $\alpha \to \infty$. *HINT: Use the eigenvalue decomposition to write* $A = \sigma_1 \vec{u}_1 \vec{u}_1^\top + \sigma_2 \vec{u}_2 \vec{u}_2^\top$ *and express* $\vec{w}$ *in the basis formed by* $\vec{u}_1, \vec{u}_2$.

Solution: Let $\vec{w} = \begin{bmatrix} 1 & 0 \end{bmatrix}^\top$. We first expand $f(\alpha\vec{w})$ as follows:

$$f(\alpha\vec{w}) = (\vec{w}^\top A\vec{w})\alpha^2 - 2b_1\alpha + c. \tag{32}$$

Note that since A is PSD, $\vec{w}^\top A\vec{w} \geq 0$. Therefore, for $f(\alpha\vec{w})$ to tend to $-\infty$ as $\alpha \to \infty$, we must have $\vec{w}^\top A\vec{w} = 0$ and $b_1 > 0$.

Using the spectral theorem since A is symmetric positive semidefinite it can be written as $A = \sigma_1 \vec{u}_1 \vec{u}_1^\top + \sigma_2 \vec{u}_2 \vec{u}_2^\top$ with $\sigma_1 \geq \sigma_2 \geq 0$ and $\vec{u}_1, \vec{u}_2$ orthonormal vectors that form a basis for $\mathbb{R}^2$. Further $\sigma_1 > 0$ since A is not the zero matrix.

Thus we can write $\vec{w} = \beta_1\vec{u}_1 + \beta_2\vec{u}_2$. Substituting this we have, $\vec{w}^\top A\vec{w} = \sigma_1\beta_1^2 + \sigma_2\beta_2^2$. For this to be zero, we must have $\beta_1 = 0$ since $\sigma_1 > 0$.

This implies $\vec{w} = \beta_2\vec{u}_2$ and we must have $\vec{u}_2 = \pm\vec{w} = \pm[1, 0]^\top$ and $\beta_2 = \pm 1$ since both $\vec{w}$ and $\vec{u}$ are unit norm. Using the fact that $\beta_2^2 = 1$ we require $\sigma_2 = 0$ for $\vec{w}^\top A\vec{w}$ to be zero. Further since $\vec{u}_1$ and $\vec{u}_2$ are orthonormal we have $\vec{u}_1 = \pm[0, 1]^\top$.

Putting everything together we can construct one example of A and b as $A = (1)\begin{bmatrix} 0 \\ 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, and $\vec{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

(d) Say that we have the data set $\{(\vec{x}_i, y_i)\}_{i=1,\ldots,n}$ of data points $\vec{x}_i \in \mathbb{R}^d$ and values $y_i \in \mathbb{R}$. Define $X = \begin{bmatrix} \vec{x}_1 & \ldots & \vec{x}_n \end{bmatrix}^\top$ and $\vec{y} = \begin{bmatrix} y_1 & \ldots & y_n \end{bmatrix}^\top$. In terms of X and $\vec{y}$, find a matrix A, a vector $\vec{b} \in \mathbb{R}^d$ and a scalar c, so that we can express the sum of the square losses $\sum_{i=1}^n (\vec{w}^\top\vec{x}_i - y_i)^2$ as the quadratic function $f(\vec{w}) = \vec{w}^\top A\vec{w} - 2\vec{b}^\top\vec{w} + c$.

Solution:

$$\sum_{i=1}^n (\vec{w}^\top\vec{x}_i - y_i)^2 = \sum_{i=1}^n \left(\vec{w}^\top\vec{x}_i(\vec{x}_i)^\top\vec{w} - 2\vec{w}^\top(y_i\vec{x}_i) + (y_i)^2\right) \tag{33}$$

Rearranging terms, we have

$$A = \sum_{i=1}^n \vec{x}_i\,(\vec{x}_i)^\top = X^\top X, \quad \vec{b} = \sum_{i=1}^n y_i\vec{x}_i = X^\top\vec{y}, \quad c = \sum_{i=1}^n (y_i)^2 = \vec{y}^\top\vec{y} = \|\vec{y}\|_2^2. \tag{34}$$

(e) Here are three statements with regards to the minimization of a quadratic loss function:

i. It can have a unique minimizer.

ii. It can have infinitely many minimizers.

iii. It can be unbounded from below, i.e. there is some direction, $\vec{w}$ so that $f(\alpha\vec{w})$ goes to $-\infty$ as $\alpha \to \infty$.

All three statements apply to general minimization of a quadratic cost function. Parts (a), (b) and (c) give concrete examples of quadratic cost functions where (i), (ii) and (iii) apply respectively. However, notice that statement (iii) cannot apply to the least squares problem as the objective is always positive. The

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least-squares problem can have infinitely many minimizers though. How? Consider the gradient of the least squares problem in part (d) at an optimal solution $\vec{w}^\star$:

$$\nabla f(\vec{w}^\star) = 2X^\top X\vec{w}^\star - 2\vec{b} = 0. \tag{35}$$

Therefore, the least squares problem only has multiple solutions if $X^\top X$ is not full rank. This means that $\operatorname{rank}\left(X^\top X\right) = \operatorname{rank}(X) < d$. Finally, the rank of X is less than d when the data points $\{\vec{x}_i\}_{i=1}^n$ do not span $\mathbb{R}^d$. This can happen when the number of data points n is less than d or when $\{\vec{f}_i\}_{i=1}^d$ are linearly dependent where $\vec{f}_i$ are the columns of X, i.e., the features.

We will see soon that these cases correspond to the *convexity* of the function: if the function is strictly convex, then it has a unique minimizer; and if it is just convex, then it can have multiple minimizers; and in both cases, it can have no minimizers. We will see soon how to prove that the quadratic objective functions we discuss in this problem are convex, strictly convex, or even non-convex.

Indicate below that you have read and understood the discussion above.

Solution: Any answer is fine.

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5. Direction of Steepest Ascent

For a differentiable function $f : \mathbb{R}^n \mapsto \mathbb{R}$ we want to show that the gradient $\nabla f(\vec{x})$ is the direction of steepest ascent at the point $\vec{x}$.

(a) Let us define the rate of change of the function $f(\vec{x})$ at the point $\vec{x}$ along an arbitrary unit vector $\vec{u}$ as:

$$D_{\vec{u}} f(\vec{x}) = \lim_{h \to 0} \frac{f(\vec{x} + h\vec{u}) - f(\vec{x})}{h}. \tag{36}$$

We call this the directional derivative. Show that the directional derivative can be equivalently expressed as $D_{\vec{u}} f(\vec{x}) = \vec{u}^\top [\nabla f(\vec{x})]$.

HINT: Use Taylor approximation of the function around the point $\vec{x}$ and evaluate it at the point $\vec{x} + h\vec{u}$.

Solution: Using Taylor's theorem we can express the function $f(\vec{x})$ as

$$f(\vec{x} + h\vec{u}) = f(\vec{x}) + [\nabla f(\vec{x})]^\top [h\vec{u}] + o(h). \tag{37}$$

We rearrange the terms, and dividing both sides by h we get

$$\frac{f(\vec{x} + h\vec{u}) - f(\vec{x})}{h} = [\nabla f(\vec{x})]^\top [\vec{u}] + \frac{o(h)}{h}. \tag{38}$$

Now we take the limit of both sides as $h \to 0$; we get

$$\lim_{h \to 0} \frac{f(\vec{x} + h\vec{u}) - f(\vec{x})}{h} = [\nabla f(\vec{x})]^\top [\vec{u}] + \lim_{h \to 0} \left(\frac{o(h)}{h} \right) \tag{39}$$

$$= [\nabla f(\vec{x})]^\top [\vec{u}]. \tag{40}$$

Note that $\lim_{h \to 0} \frac{o(h)}{h} = 0$ because $o(h)$ decays faster than h as $h \to 0$.

(b) Show that

$$\frac{\nabla f(\vec{x})}{\|\nabla f(\vec{x})\|_2} = \underset{\|\vec{u}\|_2 = 1}{\operatorname{argmax}}\, \vec{u}^\top [\nabla f(\vec{x})]. \tag{41}$$

Solution: Using Cauchy-Schwarz inequality we can write:

$$\vec{u}^\top [\nabla f(\vec{x})] \leq \|\vec{u}\|_2 \|\nabla f(\vec{x})\|_2 \tag{42}$$

$$= \|\nabla f(\vec{x})\|_2, \tag{43}$$

so the maximum value that the expression can take is $\|\nabla f(\vec{x})\|_2$. Now it remains to show that this value is attained for the choice $\vec{u} = \frac{\nabla f(\vec{x})}{\|\nabla f(\vec{x})\|_2}$.

$$\frac{\left[\nabla f(\vec{x})\right]^\top}{\|\nabla f(\vec{x})\|_2} \nabla f(\vec{x}) = \frac{\|\nabla f(\vec{x})\|_2^2}{\|\nabla f(\vec{x})\|_2} \tag{44}$$

$$= \|\nabla f(\vec{x})\|_2. \tag{45}$$

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6. Gradient Descent Algorithm

Given a continuous and differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, the gradient of f at any point $\vec{x}$, $\nabla f(\vec{x})$, is orthogonal to the level curve of f at point $\vec{x}$, and it points in the increasing direction of f. In other words, moving from point $\vec{x}$ in the direction $\nabla f(\vec{x})$ leads to an increase in the value of f, while moving in the direction of $-\nabla f(\vec{x})$ decreases the value of f. This idea gives an iterative algorithm to minimize the function f: the gradient descent algorithm.

(a) Consider $f(x) = \frac{1}{2}(x-2)^2$, and assume that we use the gradient descent algorithm:

$$x_{k+1} = x_k - \eta \nabla f(x_k) \quad \forall k \geq 0, \tag{46}$$

with some random initialization x_0, where $\eta > 0$ is the step size (or the learning rate) of the algorithm. Write $(x_k - 2)$ in terms of $(x_0 - 2)$, and show that x_k converges to 2, which is the unique minimizer of f, when $\eta = 0.2$.

Solution: For the given function, we have $\nabla f(x) = (x-2)$; therefore, the gradient descent algorithm gives

$$x_{k+1} = x_k - \eta(x_k - 2). \tag{47}$$

By subtracting 2 from both sides, we obtain

$$(x_{k+1} - 2) = (1-\eta)(x_k - 2) \implies (x_k - 2) = (1-\eta)^k (x_0 - 2). \tag{48}$$

Given $\eta = 0.2$, we have

$$|x_k - 2| = 0.8^k \, |x_0 - 2| \to 0 \quad \text{as } k \to \infty, \tag{49}$$

which shows that x_k converges to 2.

(b) Let $\alpha, \beta \in \mathbb{R}$ and that for all η such that $\alpha < \eta < \beta$, the gradient descent algorithm converges to 2 from all possible initializations in $\mathbb{R}$. What are the smallest α and the largest β? After you determine α, β, answer the following. What happens when we set (i) $\eta = \alpha$ (ii) $\eta = \beta$ (iii) $\eta > \beta$?

Solution: From the solution for part (a), we have

$$|x_k - 2| = |1-\eta|^k \, |x_0 - 2| \quad \forall k \in \mathbb{N}. \tag{50}$$

For convergence of the algorithm for every initialization, it is necessary and sufficient to have $|1-\eta| < 1$, which is equivalent to $\eta \in (0, 2)$. If $\eta = 2$, x_k oscillates around 2. If $\eta = 0$, x_k remains stationary at x_0. In both cases, $|x_k - 2|$ remains fixed. If $\eta > 2$, x_k oscillates around 2 while $|x_k|$ grows unboundedly.

(c) Now assume that we use the gradient descent algorithm to minimize $f(\vec{x}) = \frac{1}{2}\left\|A\vec{x} - \vec{b}\right\|_2^2$ for some $A \in \mathbb{R}^{m \times n}$ and $\vec{b} \in \mathbb{R}^m$, where A has full column rank. First compute $\nabla f(\vec{x})$. Note that $(A^\top A)^{-1} A^\top \vec{b}$ is the solution to the least-squares problem, and $(\vec{x}_k - (A^\top A)^{-1} A^\top \vec{b})$ is the distance from the solution at time k. Write $(\vec{x}_k - (A^\top A)^{-1} A^\top \vec{b})$ in terms of $(\vec{x}_0 - (A^\top A)^{-1} A^\top b)$.

Solution: We can write $f(\vec{x}) = \frac{1}{2}(\vec{x}^\top A^\top A \vec{x} - \vec{x}^\top A^\top \vec{b} - \vec{b}^\top A \vec{x} + \vec{b}^\top \vec{b})$, so

$$\nabla f(\vec{x}) = A^\top A \vec{x} - A^\top \vec{b}. \tag{51}$$

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Then the gradient descent algorithm gives

$$\vec{x}_{k+1} = \vec{x}_k - \eta\left(A^\top A\vec{x}_k - A^\top b\right) = \vec{x}_k - \eta A^\top A\left(\vec{x}_k - (A^\top A)^{-1}A^\top\vec{b}\right). \tag{52}$$

By subtracting $(A^\top A)^{-1}A^\top\vec{b}$ from both sides, we obtain

$$\left(\vec{x}_{k+1} - (A^\top A)^{-1}A^\top\vec{b}\right) = \left(I - \eta A^\top A\right)\left(\vec{x}_k - (A^\top A)^{-1}A^\top\vec{b}\right) \tag{53}$$

and consequently,

$$\left(\vec{x}_k - (A^\top A)^{-1}A^\top\vec{b}\right) = \left(I - \eta A^\top A\right)^k\left(\vec{x}_0 - (A^\top A)^{-1}A^\top\vec{b}\right). \tag{54}$$

(d) Now consider $f(\vec{x}) = \frac{1}{2}\left\|A\vec{x} - \vec{b}\right\|_2^2 + \frac{1}{2}\lambda\left\|\vec{x}\right\|_2^2$ for some $A \in \mathbb{R}^{m\times n}$, $\vec{b} \in \mathbb{R}^m$, and $\lambda \in \mathbb{R}$, where A has full column rank and $\lambda \geq 0$. Suppose we solve this problem via gradient descent with step-size $\eta = \frac{1}{\sigma_1^2+\lambda}$, where σ_1 is the maximum singular value of A. Show the gradient descent converges.

Solution: We can write $f(\vec{x}) = \frac{1}{2}(\vec{x}^\top A^\top A\vec{x} - \vec{x}^\top A^\top\vec{b} - \vec{b}^\top A\vec{x} + \vec{b}^\top\vec{b} + \lambda\vec{x}^\top\vec{x})$, so

$$\nabla f(\vec{x}) = A^\top A\vec{x} - A^\top\vec{b} + \lambda\vec{x} = (A^\top A + \lambda I)\vec{x} - A^\top\vec{b}$$

The least-squares solution to this problem is now $x^\star = (A^\top A + \lambda I)^{-1}A^\top\vec{b}$. Let's consider the distance from $x^\star$ at time k+1

$$\left(\vec{x}_{k+1} - (A^\top A + \lambda I)^{-1}A^\top\vec{b}\right) = \left(I - \eta(A^\top A + \lambda I)\right)\left(\vec{x}_k - (A^\top A + \lambda I)^{-1}A^\top\vec{b}\right) \tag{55}$$

and consequently,

$$\left(\vec{x}_k - (A^\top A + \lambda I)^{-1}A^\top\vec{b}\right) = \left(I - \eta(A^\top A + \lambda I)\right)^k\left(\vec{x}_0 - (A^\top A + \lambda I)^{-1}A^\top\vec{b}\right). \tag{56}$$

For the algorithm to converge, we need the largest eigenvalue of $(I - \eta(A^\top A + \lambda I))$ to be less than 1 in absolute value. We know any eigenvector v_i of $A^\top A$ is also eigenvector of $I - \eta(A^\top A + \lambda I)$ with eigenvalue $1 - \eta(\sigma_i^2 + \lambda)$. With $\eta = \frac{1}{\sigma_1^2+\lambda}$, where σ_1 is the maximum singular value of A given, the following inequality will always be true

$$\forall i,\ |1 - \eta(\sigma_i^2 + \lambda)| < 1. \tag{57}$$

Therefore, the gradient descent eventually converges to the least-squares solution $x^\star$.

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7. Homework Process

With whom did you work on this homework? List the names and SIDs of your group members.

NOTE: If you didn't work with anyone, you can put "none" as your answer.

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